

Qatar-1b

R-1219

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-1_201021 · created 2026-07-31T08:37:33+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2459143.73 +/- 0.0019 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.083 +/- 0.033
Transit depth (Rp/Rs)^2	0.7 +/- 0.55 [%]
Orbital Inclination (inc)	84.85 +/- 2.61
Airmass coefficient 1 (a1)	0.929 +/- 0.012
Airmass coefficient 2 (a2)	0.0443 +/- 0.0096
Scatter in the residuals of the lightcurve fit is	1.29 %
Best Comparison Star	#3 - [427, 188]
Optimal Method	PSF photometry
Transit Duration (day)	0.062 +/- 0.017

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.083 ± 0.033 vs 0.1463 (deviation 1.92σ)
Duration fit vs published	0.062 d vs 0.0692 d (deviation 0.42σ)
Mid-time O–C	-4.6 min
Depth SNR	2.5
Residual scatter	1.29 %

Light curve

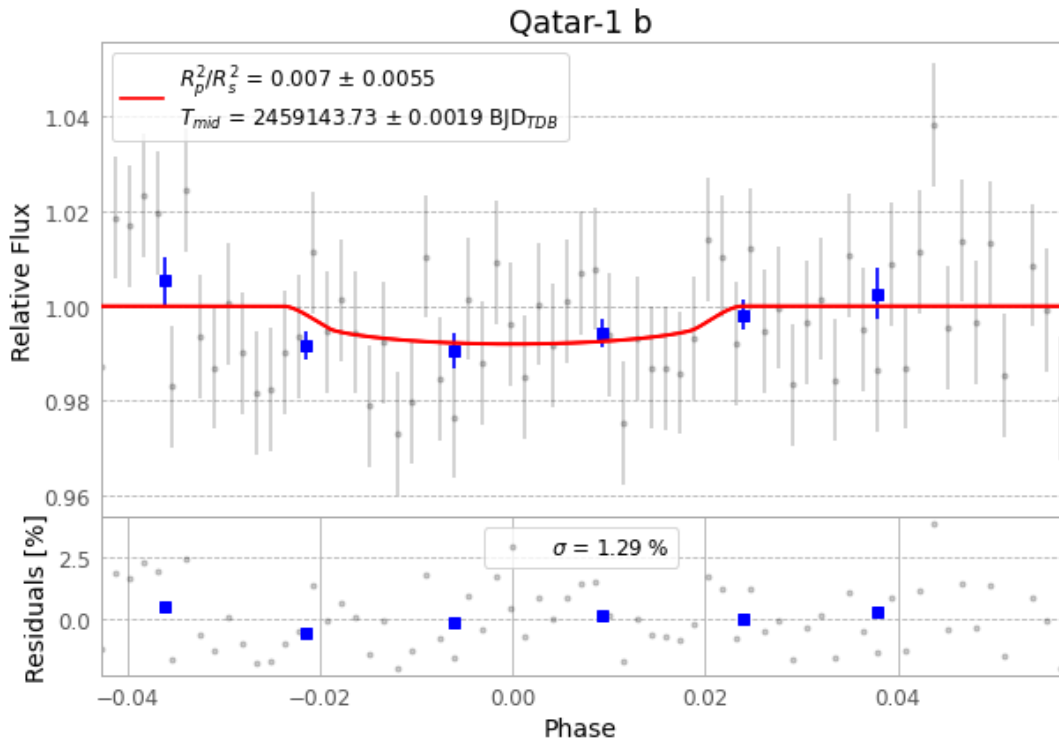


Figure 1 — FinalLightCurve_Qatar-1 b_2020-10-20.png (SHA-256 27798c4f4dbb963c...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [449, 343], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 239ef64958a407e0...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

69 FITS frames (45 MB), Qatar-1201021040012.FITS → Qatar-1201021072412.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-201021.log (3b284318158e...), FinalLightCurve_Qatar-1 b_2020-10-20.png (27798c4f4dbb...), FinalLightCurve_Qatar-1 b_2020-10-20.csv (e9e11f7efac7...)

Compute resources

Wall time 120.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-31 08:37Z.